

Object Oriented Programming Lab File



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Experiment 1: Introduction to Java Environment

Answer The Following and submit it as a part of lab experiment work:

1. Explore and understand the role of JDK, JRE, and JVM
2. Install the latest available JDK and verify the Java Environment.
3. Create a Sample Hello World Program using a simple text editor (e.g., Notepad) and show the steps to compile and execute the program using the command prompt.
4. Display your name and complete address in different lines.

Answer the following:

1. Explore and understand the role of JDK, JRE, and JVM

Ans:

These three components are essential for running and developing Java applications.

JVM (Java Virtual Machine)

- JVM is responsible for running Java programs.
- It converts Java bytecode (compiled code) into machine code that the operating system understands.
- It provides memory management, garbage collection, and security.
- JVM is platform-dependent but executes platform-independent bytecode.

JRE (Java Runtime Environment)

- JRE contains JVM and essential libraries to run Java applications.
- It does *not* include development tools like the compiler.
- If you only want to run Java programs (not develop them), installing JRE is enough.

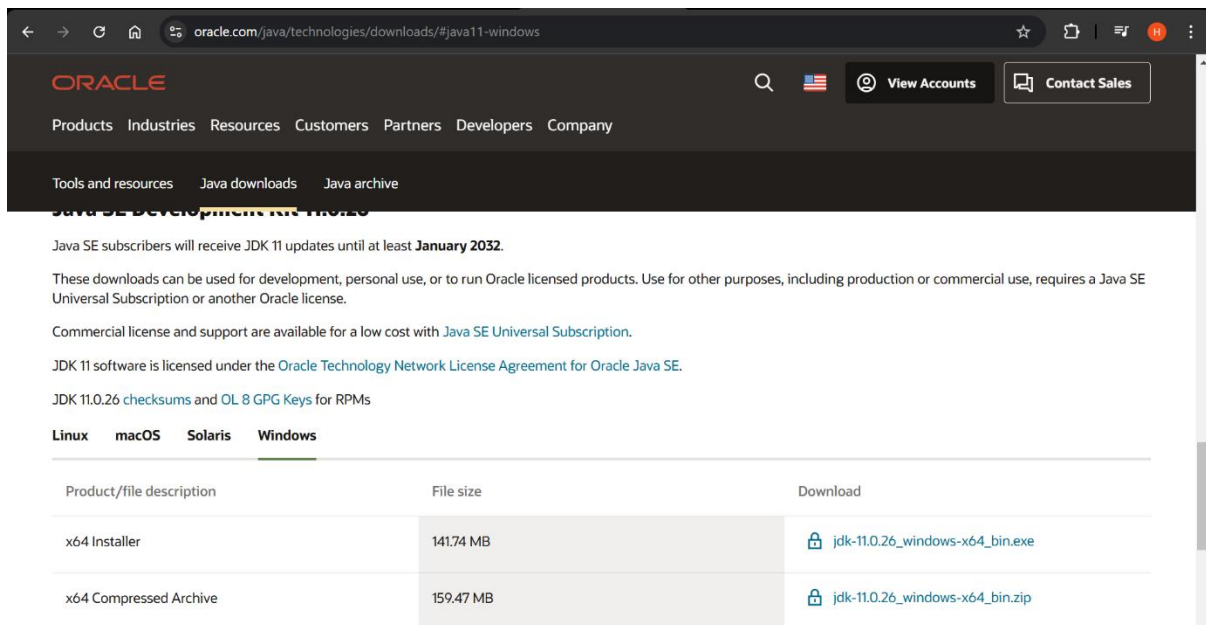
JDK (Java Development Kit)

- JDK includes JRE + additional tools for development (compiler, debugger, etc.).
- It is required for writing, compiling, and running Java programs.
- It contains javac (Java compiler) and java (Java runtime).

2. Install the latest available JDK and verify the Java Environment.

Ans:

Installation of latest version of JDK:



The screenshot shows the Oracle Java Downloads page for Windows. The page includes a navigation bar with the Oracle logo, a search icon, a US flag, and buttons for 'View Accounts' and 'Contact Sales'. Below the navigation bar, there are links for 'Products', 'Industries', 'Resources', 'Customers', 'Partners', 'Developers', and 'Company'. A secondary navigation bar contains 'Tools and resources', 'Java downloads', and 'Java archive'. The main content area features a heading 'Java SE Development Kit 11.0.26' and a subheading 'Java SE Development Kit 11.0.26'. It includes a notice that Java SE subscribers will receive JDK 11 updates until at least January 2032. Below this, there is a paragraph stating that these downloads can be used for development, personal use, or to run Oracle licensed products. A link for 'Java SE Universal Subscription' is provided. The page also mentions that JDK 11 software is licensed under the Oracle Technology Network License Agreement for Oracle Java SE. A link for 'JDK 11.0.26 checksums and OL 8 GPG Keys for RPMs' is also present. At the bottom, there is a table with columns for 'Linux', 'macOS', 'Solaris', and 'Windows'. The 'Windows' column is selected, and it contains a table with two rows: 'x64 Installer' and 'x64 Compressed Archive'. Each row lists the product/file description, file size, and a download link.

Linux	macOS	Solaris	Windows

Product/file description	File size	Download
x64 Installer	141.74 MB	jdk-11.0.26_windows-x64_bin.exe
x64 Compressed Archive	159.47 MB	jdk-11.0.26_windows-x64_bin.zip

3. Create a Sample Hello World Program using a simple text editor (e.g., Notepad) and show the steps to compile and execute the program using the command prompt.

Ans:

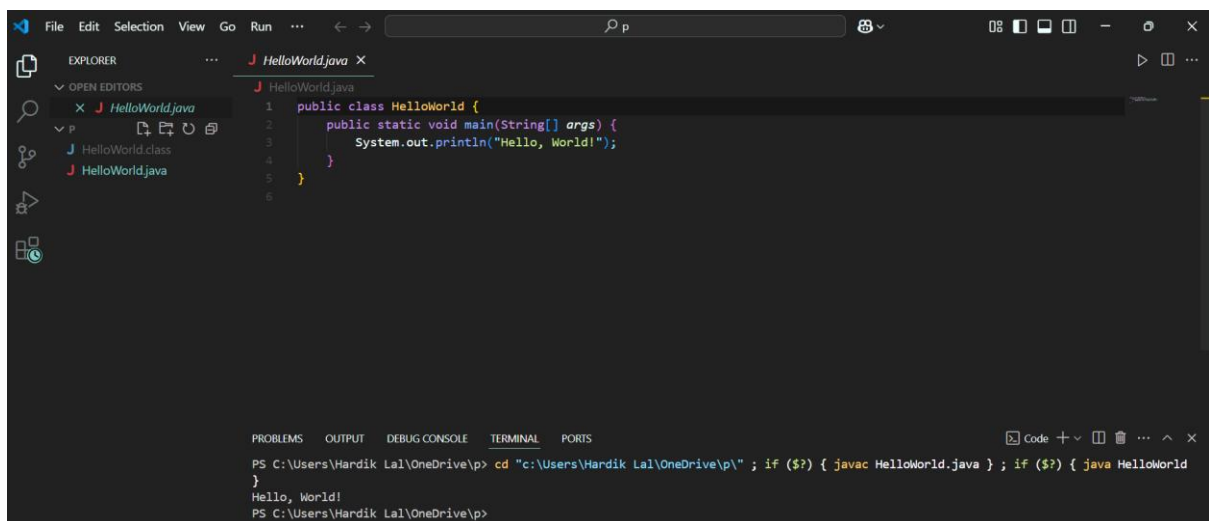
Write the Code in any text editor like notepad



A screenshot of a Notepad window titled 'HelloWorld.java'. The window has a menu bar with 'File', 'Edit', and 'View'. The code inside is as follows:

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

We can run the code in any compiler

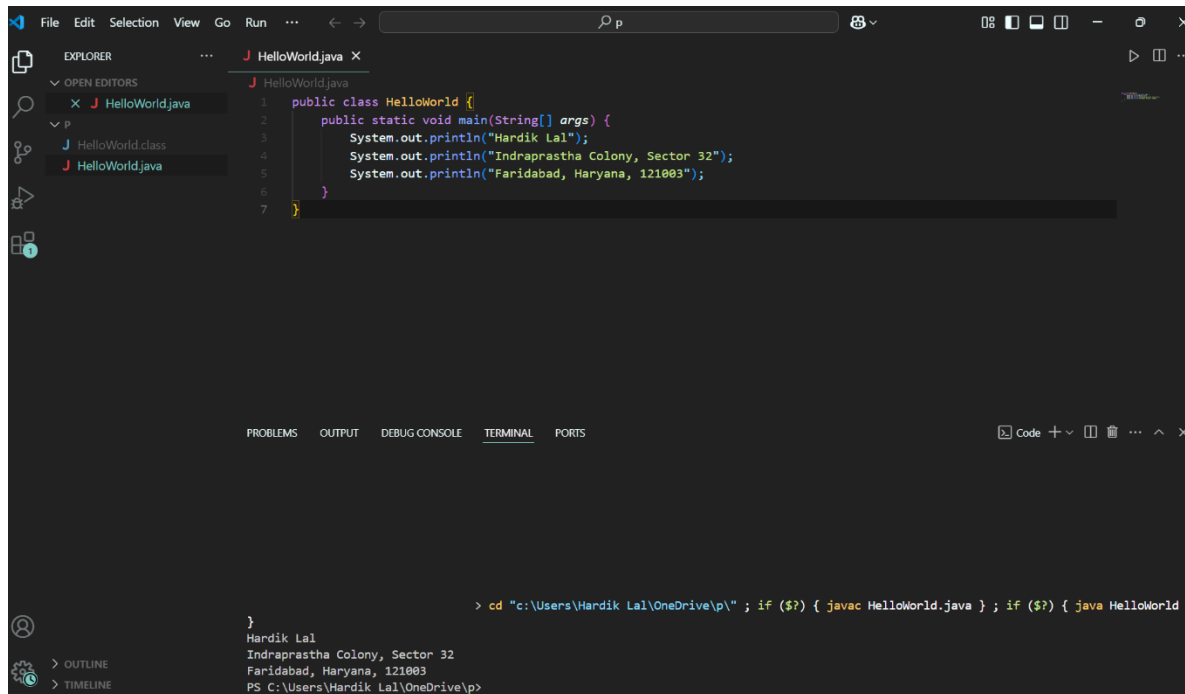


A screenshot of an IDE (Visual Studio Code) showing the 'HelloWorld.java' file. The Explorer panel on the left shows the file structure. The main editor shows the code. The terminal at the bottom shows the command prompt output:

```
PS C:\Users\Hardik Lal\OneDrive\p> cd "c:\Users\Hardik Lal\OneDrive\p" ; if ($?) { javac HelloWorld.java } ; if ($?) { java HelloWorld  
}  
Hello, World!  
PS C:\Users\Hardik Lal\OneDrive\p>
```

4. Display your name and complete address in different lines.

Ans:



The screenshot shows an IDE with a Java file named `HelloWorld.java`. The code defines a `HelloWorld` class with a `main` method that prints the name and address on three separate lines. The terminal output shows the program being executed and the corresponding printed text.

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hardik Lal");  
        System.out.println("Indraprastha Colony, Sector 32");  
        System.out.println("Faridabad, Haryana, 121003");  
    }  
}
```

```
> cd "c:\Users\Hardik Lal\OneDrive\p\" ; if ($?) { javac HelloWorld.java } ; if ($?) { java HelloWorld  
}  
Hardik Lal  
Indraprastha Colony, Sector 32  
Faridabad, Haryana, 121003  
PS C:\Users\Hardik Lal\OneDrive\p>
```